

AMENDMENT NO. 1 SEPTEMBER 1994
TO
IS 4321 : 1989 PESTICIDES — 2, 4-D, TECHNICAL —
SPECIFICATION
(Second Revision)

(Page 2, Annex B) — Substitute the following for the existing:

‘DETERMINATION OF FREE PHENOLS IN 2, 4-D

B-1 OUTLINE OF THE PROCEDURE

B-1.1 The absorbance of acetone-ammonia solution of the acid is determined after adding 4-aminophenazone and potassium ferricyanide solution and comparing the colour development with the similarly prepared solution of standard 2, 4-dichlorophenol solution.

B-2 REAGENTS

B-2.1 Standard 2, 4-Dichlorophenol Solution

Prepare by dissolving 100 mg. of pure 2, 4-dichlorophenol in 10 ml of acetone and diluting to 500 ml with distilled water. Dilute 10 ml of the resulting solution to 100 ml with distilled water. One millilitre of the final solution is equivalent to 20 µg of 2, 4-dichlorophenol.

B-2.2 Standard 4-Aminophenazone Solution — 2 percent (m/v), in distilled water and to be stored in dark.

B-2.3 Standard Potassium Ferricyanide Solution — 8 percent (m/v), freshly prepared in distilled water.

B-2.4 Standard Ammonium Hydroxide Solution — 0.1N.

B-3 PROCEDURE AND CALIBRATION

B-3.1 Dissolve 200 mg of the sample in 60 ml of standard ammonium hydroxide solution, dilute to 500 ml with distilled water, and take with a pipette 50 ml of the resulting solution into a conical flask. Add 0.5 ml of the standard 4-aminophenazone solution, shake, add 0.25 ml of the potassium ferricyanide solution, and shake again for one minute.

B-3.2 Pipette 10 ml of the 2, 4-dichlorophenol standard solution into a conical flask, add 5 ml of standard ammonium hydroxide solution and dilute to 50 ml with distilled water. Mix thoroughly, add 0.5 ml of the standard 4-aminophenazone

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solution, shake, and add 0.25 ml of potassium ferricyanide solution, and shake again for one minute. Similarly, pipette out 2, 4, 6, 8 and 12 ml of the standard solution and repeat the procedure. Plot a graph of optical density against concentration and check for linearity.

B-3.3 Measure the colour intensity with the help of spectrophotometer at 510 nm and compare the optical density of the sample with that of the standard and calculate the free phenol (as 2, 4-dichlorophenol) content.'

(FAD 1)

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